CS6230 CAD for VLSI: Physical Design Report (Synthesis and Placement)

Roll no.: EE26E016 (Kishore) • Partner: Rishi (EE25S082) • 9 November 2026 • *mkQRDecoder*, Bluespec, SAED 14nm RVT

1 Flow and constraints

Synthesis used Synopsys Design Compiler (U-2022.12) with SAED 14nm RVT, normal synthesis (`compile_ultra`, no topographical mode) and external input and output delays of 40 % of the clock period in every SDC. The best netlist was placed with Synopsys IC Compiler (V-2023.12, ICC, not ICC2) using the same libraries and constraints: floorplan, power straps and PlaceOpt. Division of work: **Kishore** ran all DC and ICC flows and wrote the scripts in `pd/`; **Rishi** explored the four-bank R5 variant. Both used the same RTL, which is identical in both submissions.

2 Post-synthesis results (Table 5)

Design number	Critical path clk period (ns)	Critical path slack (ns)	Max clock freq (MHz)	Total number of leaf cells	Total number of sequential cells	Total power (mW)	Design area (µm²)
1	5.00	0.00	200.0	115,224	32,102	9.9178	216,341.52
2	5.00	0.00	200.0	90,815	36,247	4.5249	141,711.74
3	2.50	0.00	400.0	92,463	35,744	8.8798	140,112.70
4	2.00	0.00	500.0	89,407	36,657	11.9537	151,986.89
5 (best)	1.20	0.00	833.3	92,233	36,667	19.8907	143,448.02

Table 5: Post-synthesis QoR (Design Compiler). Max clock frequency = 1000 / period. Slack is the worst slack of all path groups (TNS 0.00 in CLK, f2f, i2f, f2o).

From design 1 to design 5 the clock period fell by 76 % (5.0 → 1.2 ns) and the design area by 33.7 % (216,341.52 → 143,448.02 µm²). Power rises with frequency (4.52 mW at 5.0 ns to 19.89 mW at 1.2 ns); at the identical 5.0 ns, design 2 uses 54.4 % less power than design 1.

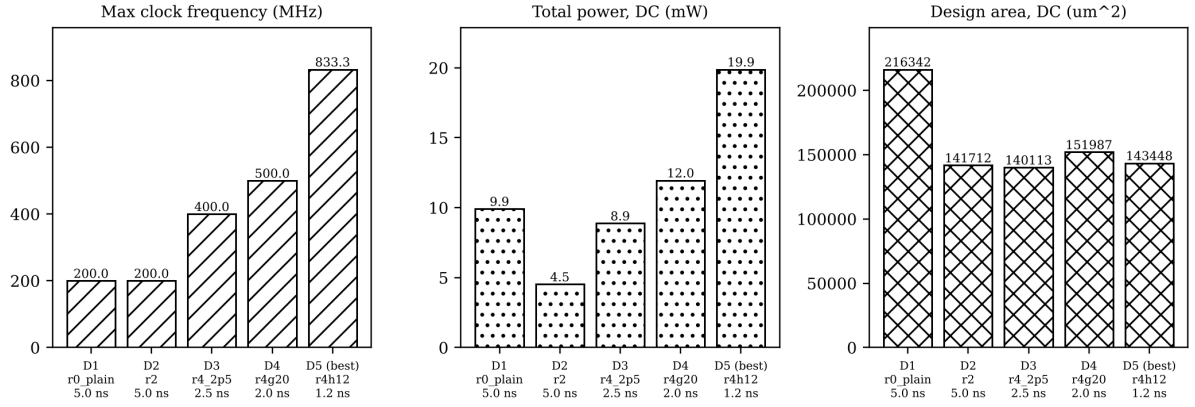


Figure 1: Frequency, power and design area of the five synthesis runs.

3 Design → synthesis → analyse → re-design

Iteration	What the reports showed	Change applied	Result
1 r0_plain	Single-cycle BM loop, un-gated 29.6 kbit store; 158 logic levels	baseline	5.00 ns met; 9.9178 mW
2 r2	Power dominated by un-gated register clocks and store muxes	BM in 3 phases, clock gating on, same 5.0 ns	power −54.4 %; area −34.5 %
3 r4_2p5	BM and store access now limit the clock	2-stage BM, 1-cycle store pipeline	2.5 ns met

Iteration	What the reports showed	Change applied	Result
4 r4g20	Dividers in front end and payload; cells with pin-access problems	reciprocal multiply-shift, pipelined BCH and zigzag, dont_use cells	2.0 ns met; 17 logic levels
5 r4h12 (best)	Design 4 closes with margin in all groups	same architecture, timing fixes, clock tightened	1.2 ns met, f2f 1.19 ns; 4.17× faster than 1

Table 3: Iteration log with the quantified improvement.

4 Post-placement results of the best design (Table 6)

Design number	Critical path clk period (ns)	Critical path slack	Max clock freq (MHz)	Total number of leaf cells	Total number of sequential cells	Total power (mW)	Block dimensions (µm)	Design area (µm²)
1 (best)	1.20	0 violating paths (f2f path 0.80 ns)	833.3	85,490	36,667	7.3985	243.98 × 243.45	52,285.97

Table 6: Post-place QoR (ICC PlaceOpt, run pn_r4h12_L244). Design area = placed standard-cell area.

- **Power:** internal 6.3528 + switching 0.9290 + leakage 0.1167 = **7.3985 mW** (`report_power`).
- **Utilisation:** 92.29 % against the ≥ 70 % requirement; 243.98 × 243.45 µm block; 0 violating setup and hold paths.
- **Against synthesis:** leaf cells 92,233 → 85,490, power 19.89 → 7.40 mW, sequential cells unchanged at 36,667.

5 Core-size sweep and alternatives

Of 17 PlaceOpt runs on the server, the five below place cleanly. The 1.10 ns netlist also places clean but gives no area or power benefit over 1.2 ns; the four-bank R5 variant (0.55–0.70 ns, 12 runs) left 1–14 violating paths in every run and was not selected.

Netlist	Core (µm)	Violating paths	f2f path (ns)	Cell area (µm²)	Utilisation (%)
r4h11 (1.10 ns)	242	0	0.76	52,636.24	94.41
r4h11 (1.10 ns)	244	0	0.75	52,626.12	92.89
r4h12 (1.20 ns)	242	0	0.82	52,269.68	93.75
r4h12 (1.20 ns), submitted	244	0	0.80	52,285.97	92.29
r4h12 (1.20 ns)	246	0	0.81	52,262.04	90.57

Table 4: Clean placement runs.

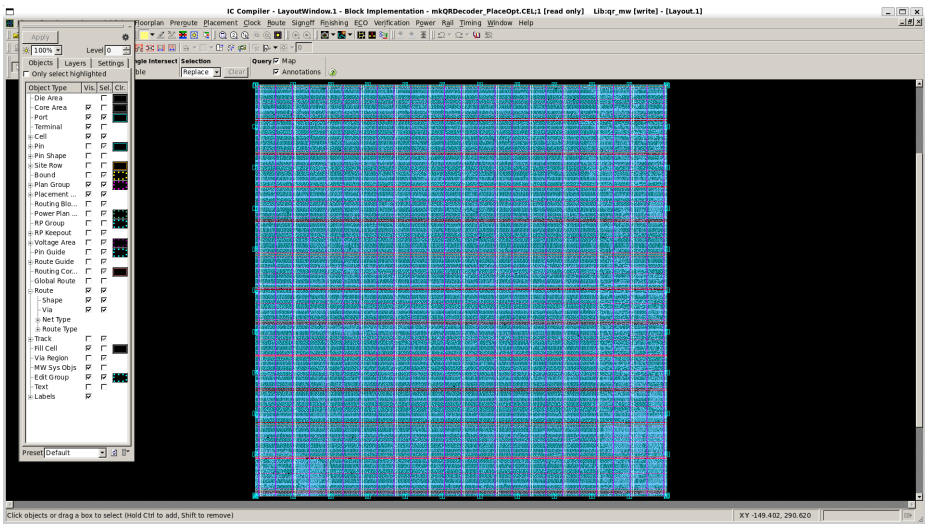


Figure 2: Placed design of the submitted run in the ICC layout window.

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Violating|Design \(Hold\)|Total Hold|Total Negative" pd
/best_design/best_design_post_place_qor.rpt | head -40
13:  Critical Path Length:          0.80
14:  Critical Path Slack:           0.00
16:  Total Negative Slack:          0.00
17:  No. of Violating Paths:        0.00
19:  Total Hold Violation:          0.00
26:  Critical Path Length:          0.62
27:  Critical Path Slack:           0.10
29:  Total Negative Slack:          0.00
30:  No. of Violating Paths:        0.00
32:  Total Hold Violation:          0.00
39:  Critical Path Length:          0.45
40:  Critical Path Slack:           0.00
42:  Total Negative Slack:          0.00
43:  No. of Violating Paths:        0.00
45:  Total Hold Violation:          0.00
54:  Leaf Cell Count:              85490
60:  Sequential Cell Count:        36667
77:  Cell Area:              52285.972357
105:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
108:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
126:Std cell area:        1177619  sites, (non-fixed:1177619 fixed:0)
128:Macro cell area:      0        sites
202:Total cell area:                52285.972357
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3: ICC post-place QoR report of the submitted run.

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; cat /scratch/CS6230_EE26E016/qr/work/pd/icc/r
un_pn_r4h12_L244/reports/PlaceOpt_Util | head -30

****************************************
  Report : Chip Summary
  Design : mkQRDecoder
  Version: V-2023.12
  Date   : Wed Nov  4 20:33:41 2026
****************************************
Std cell utilization: 92.29%  (1177619/(1275939-0))
(Non-fixed + Fixed)
Std cell utilization: 92.29%  (1177619/(1275939-0))
(Non-fixed only)
Chip area:            1275939  sites, bbox (0.00 0.00 243.98 243.45) um
Std cell area:        1177619  sites, (non-fixed:1177619 fixed:0)
                      85490    cells, (non-fixed:85490  fixed:0)
Macro cell area:      0        sites
                      0        cells
Placement blockages:  0        sites, (excluding fixed std cells)
                      0        sites, (include fixed std cells & chimney area)
                      0        sites, (complete p/g net blockages)
Routing blockages:    0        sites, (partial p/g net blockages)
                      0        sites, (routing blockages and signal pre-route)
Lib cell count:       186
Avg. std cell width:  0.95 um
Site array:           unit     (width: 0.074 um, height: 0.600 um, rows: 387)
Physical DB scale:    1000 db_unit = 1 um


****************************************
  Report : pnet options
  Design : mkQRDecoder
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 4: Utilisation report of the submitted run.

6 Notes and limitations

- Design Compiler's `report_constraint` lists `max_area` as violated only because the constraint is `set_max_area 0` (an area goal); it is not a timing violation, and all timing groups have TNS 0.00.
- The physical flow stops at placement. Routing, DRC and LVS were not run, so the layout is not claimed to be DRC or LVS clean.